

JIAXU ZHOU

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EDUCATION

The Chinese University of Hong Kong B.Sc. in Mathematics and Information Engineering Minor in Physics GPA: 3.939/4.000	Sep 2023 – Jun 2027
University of California, Berkeley Visiting Student Researcher (Advisor: Prof. Koushil Sreenath)	May 2026 – Sep 2026
University of Washington Exchange Student (GPA 4.000/4.000)	Sep 2025 – Dec 2025

RESEARCH EXPERIENCE

Guiding Diffusion Motion Prior with World Model for Humanoid Control University of California, Berkeley Advised by Prof. Koushil Sreenath	May 2026 – Sep 2026
• First author, submitted to ICRA2027. • Proved classifier guidance can be viewed as optimizing a prior-regularized Model Predictive Control objective. • Trained a task-agnostic humanoid diffusion motion prior, and guided it via a learned state space world model for task-specific control. Deployed on Unitree G1 Humanoid.	
Single Forward Evaluation Flow Policy for Robotic Manipulation University of California, Berkeley Advised by Prof. Koushil Sreenath	May 2026 – Sep 2026
• Second author, submitted to ICRA2027. • Distilled multi-step flow policies into a single network via hidden-state alignment, enabling one-pass inference. • Outperformed flow-based and diffusion policies while significantly reducing inference latency and deployed on an SO-101 robot arm.	
Modeling Latent Dynamics with Port-Hamiltonian Network The Chinese University of Hong Kong Advised by Prof. Shengchao Liu	Oct 2025 – May 2026
• Modeled Transformer's token generation trajectory as a port-Hamiltonian system. Regularized and improved LLM's reasoning capability.	
Emergent Behavior in Multi-Agent Systems Remote Collaboration with Carnegie Mellon University With Prof. Maarten Sap	Mar 2025 – Sep 2025
• First author, major Revision at Nature Human Behavior. • Showed that many reported emergent behaviors in current LLM multi-agent studies arise from human-designed experimental artifacts via controlled experiments.	
Multi-Agent Collaboration under Agent Failures Remote Collaboration with Carnegie Mellon University With Prof. Maarten Sap	Nov 2023 – Mar 2025
• Second author, ICML 2025. • Studied the structural resilience of multi-agent systems when adversarial agents are involved in collaboration.	
Scalable Multi-Agent Simulation Infrastructure Remote Collaboration with Carnegie Mellon University With Prof. Maarten Sap	Nov 2024 – Jan 2025
• Fourth author; NAACL 2025 System Demonstrations; code contributor.	

- Developed infrastructure and communication protocols for flexible social simulations with 100+ concurrent LLM agents.
- Contributed to the open-source SOTOPIA platform (**335 stars**): github.com/sotopia-lab/sotopia.

Improving Spatial Understanding of Vision-Language Models

Project at University of Washington

Team Lead

Sep 2025 – Dec 2025

- Identified jigsaw solving as a fundamental spatial reasoning task and showed that current VLMs perform poorly on it.
- Fine-tuned VLMs on jigsaw reasoning and observed improved generalization across multiple visual reasoning benchmarks.

PUBLICATIONS

Jiaxu Zhou, Yiyang Shao, Yaokun Han, Yufeng Chi, Yakun Sophia Shao, Koushil Sreenath. “WMGuide: Guiding Diffusion Action Prior with Learned World Model for Humanoid Planning.” *Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.*

Yaokun Han, **Jiaxu Zhou**, Sattvik Saparam, Koushil Sreenath. “BiFlow Policy: Bidirectional Normalizing-Flow Policies for Robotic Manipulation with One Network Evaluation.” *Submitted to IEEE International Conference on Robotics and Automation (ICRA), 2027.*

Jiaxu Zhou*, Jen-Tse Huang*, Xuhui Zhou, Man Ho Lam, Xintao Wang, Hao Zhu, Wenxuan Wang, Maarten Sap. “The PIMMUR Principles: Ensuring Validity in Collective Behavior of LLM Societies.” *Major Revision at Nature Human Behavior, 2026.*

Jen-Tse Huang, **Jiaxu Zhou**, Tailin Jin, Xuhui Zhou, Zixi Chen, Wenxuan Wang, Youliang Yuan, Michael Lyu, Maarten Sap. “On the Resilience of LLM-Based Multi-Agent Collaboration with Faulty Agents.” *Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.*

Xuhui Zhou, Zhe Su, Sophie Feng, **Jiaxu Zhou**, Jen-Tse Huang, Hsien-Te Kao, Spencer Lynch, Svitlana Volkova, Tongshuang Wu, Anita Woolley, et al. “SOTOPIA-S4: A User-Friendly System for Flexible, Customizable, and Large-Scale Social Simulation.” *Proceedings of NAACL-HLT, System Demonstrations, 2025.*

HONORS AND AWARDS

- Professor Charles K. Kao Research Exchange Scholarship 2026
supported summer research to University of California, Berkeley
- Talent Development Scholarship from Hong Kong Government 2025
- Dean’s Honours List 2023, 2024
- Mr. & Mrs.Chin F. Foin Memorial Scholarship 2023,2024
awarded to the top-performing student in each major
- First Prize, National Olympiad in Informatics in Provinces (NOIp) 2021

TECHNICAL SKILLS

- **Programming:** Python, C++, PyTorch, JAX
- **Robotics and Control:** Isaac Lab, MuJoCo, ROS 2
- **Math & Physics:** optimization, probability, dynamical systems, analytical mechanics, quantum mechanics
- **Hardware & design:** analog/digital circuits, signal processing
- **Languages:** Mandarin, Cantonese, English (TOEFL 111/120)